Intermediate · AI data pipelines · Highlights

# Synthetic Data Generation for Smarter AI Workflows

Turn a handful of real examples into a large, validated training set without hand-labeling everything yourself.

For developers who want to fine-tune or ground a model on their own documents and are short on labeled data · 27 July 2026

The one- or two-page version: what matters, the topics it covers, and every term defined. Read the full lesson for the explanations and worked examples.

Watch the source video · Read the full lesson · Read the condensed version

## What matters

- ✓ Models cannot learn directly from raw documents; unstructured text, tables, and equations must first be converted into structured data.
- ✓ A small set of hand-written question-and-answer pairs, called seed data, teaches a generation model the pattern and style to imitate.
- ✓ Synthetic data generation is a pipeline, not a single API call: it generates candidate examples, transforms them, and validates them before they are used.
- ✓ Validation checks synthetic examples for faithfulness to the source, relevance to the question, and diversity across the dataset.
- ✓ Generation pipelines can run entirely inside your own environment, connecting to local or self-hosted models, so source data never has to leave it.
- ✓ Synthetic data lets you preserve privacy, balance rare classes, and augment thin domains without collecting more real-world data.
- ✓ Because the pipeline is generate-validate-deploy on repeatable steps, the resulting dataset is reproducible — a requirement for most enterprise AI workflows.

## What it covers

01 **Why you can't just point a model at a PDF** `0:00`
A model can't learn from a document directly, and even structured content usually isn't enough of it to shift a base model's weights.

02 **Extracting structured context from raw documents** `0:41`
Tools like Docling turn PDFs and scans into structured data — sections, headings, tables — using OCR and parsing, which is the first step before any generation happens.

03 **Generating synthetic data from seed examples** `1:28`
Once seed data exists, a generation model studies its patterns and produces new, similar question-and-answer pairs at scale — this is synthetic data generation.

04 **Validating synthetic data before you trust it** `2:14`
Generated examples are checked for faithfulness, relevance, and diversity, and the whole flow can run locally against your own model endpoint so source data never leaves your environment.

05 **Why control over the data matters** `2:55`
Synthetic data generation isn't just about producing more text — it gives you privacy preservation, class balancing, and a reproducible pipeline you can run before every deployment.

## Glossary

**Seed data** — A small, hand-written set of example question-and-answer pairs that demonstrates the correct pattern and style for a model to imitate when generating more examples.

**OCR (optical character recognition)** — Software that converts images of text, such as a scanned page, into machine-readable text characters.

**Docling** — An open source tool that converts PDFs and scanned documents into structured data — sections, headings, tables — using OCR and parsing.

**OpenAI-compatible endpoint** — An API that accepts requests in the same format as OpenAI's API, letting tools switch between providers (self-hosted, Claude, etc.) without changing code.

**Synthetic data** — New data examples produced by a model rather than collected from the real world, generated to follow the statistical pattern of a smaller real dataset.

**Faithfulness** — A validation check on whether a generated answer's claims are actually supported by the source document, as opposed to invented.

**SDGHub** — An open source project that runs synthetic data generation flows: pipelines that generate, transform, and validate synthetic examples.

**Fine-tuning** — Continuing to train an existing base model on a smaller, task-specific dataset so its weights shift toward that task.